

WAB

Provadis School of International Management and Technology

Examining the Feasibility of Machine Learning-Based LogP Prediction on an ESP32 Microcontroller

A Proof-of-Concept Implementation and Performance Analysis

Rubin Chempananickal James

rubin.chempananickal-james@stud-provadis-hochschule.de

Department: Information Technology

Module: Embedded Systems and Software

Reviewer: Prof. Dr. Eric Hutter

February 16, 2026

Abstract

Contents

Abstract	II
1 Introduction	1
1.1 Background	1
1.2 Research Questions	1
2 Methods	2
2.1 Design	2
2.2 Data Collection	2
AI Declaration	i
Declaration of Authorship	ii

List of Figures

List of Tables

1 Introduction

1.1 Background

1.2 Research Questions

2 Methods

2.1 Design

2.2 Data Collection

AI Declaration

The usage of AI tools within this project is documented here. I solemnly declare that I have documented all interactions with AI tools, including the prompts used and the outputs received.

System	Prompt	Usage
ChatGPT 1	Asked to delete the existing project and provide a simple LaTeX template with a shared preamble, a Chapters subdirectory, TOC, glossary, abbreviations, and Roman numerals for non-content pages.	Template structure and LaTeX setup provided
ChatGPT 2	Requested additional title page lines (WAB header, reviewer, module) for the main document.	Title page metadata and layout updated
ChatGPT 3	Requested uppercase Roman numerals for front matter and lowercase Roman numerals for back matter page numbering.	Page numbering adjusted in main and exposee
ChatGPT 4	Requested exposee title page to include shared info fields (WAB header, department, module, reviewer).	Exposee title page updated to include shared metadata
ChatGPT 5	Requested adding the provadis-hochschule.pdf logo to the top right corner of both main and exposee title pages.	Logo added to top right of both title pages
ChatGPT 6	Reported that Glossary and Abbreviations sections were missing from compiled output.	Added example <code>\newacronym</code> entries to <code>abbreviations.tex</code> so the Abbreviations section displays properly

Declaration of Authorship

I hereby confirm that I have personally and independently prepared the present work and have not used any sources or aids other than those specified. All passages taken verbatim or in substance from other sources are identified as such. The drawings, illustrations and tables in this work are created by me or have been provided with an appropriate source reference. This work has not been submitted by me to any other university in the same or similar form for the acquisition of an academic degree.

Frankfurt, February 16, 2026

Rubin Chempananickal James